

SW8112Q

6-9GHz UWB-band RF Front-End Module

Aug 2020 V1.0

DESCRIPTION

The SW8112Q is a Front-End Module (FEM) with a fully-integrated Power Amplifier (PA), Low-Noise-Amplifier (LNA) and Filter to reject out-of-band spurious. The SW8112Q requires no external matching component, reduces assembly complexity and the PCB area, enabling a cost-effective solution. The SW8112Q achieves high transmitting power, high transmit/receive switching loss. All these features make SW8112Q an excellent choice for UWB FEM as it improves sensitivity with low noise figure and better distance with high output power, meanwhile it reduces the overall cost of the UWB transceiver.

FEATURES

- Fully-Integrated PA/LNA/Pre-RX FILTER;
- Tx output Saturated Power= +18.0 dBm@7GHz;
- Tx Power Gain
23.0 dB@6.5GHz
20.0 dB@8.0GHz
- Rx Noise Figure
2.0 dB@6.5GHz
2.2 dB@8.0GHz
- Rx Power Gain
20.0 dB@6.5GHz
15.0 dB@8.0GHz
- RX Filter Rejection
>36dBc Rejection @DC~2.4GHz bands;
>18dBc Rejection @2.4~4.8GHz bands;
- Slim QFN-16L package:3.0 mm X 3.0 mm X 0.55 mm.

APPLICATIONS

- Transceivers with IEEE 802.15.4-2011 UWB compliant.

PIN CONFIGURATION AND MARKING

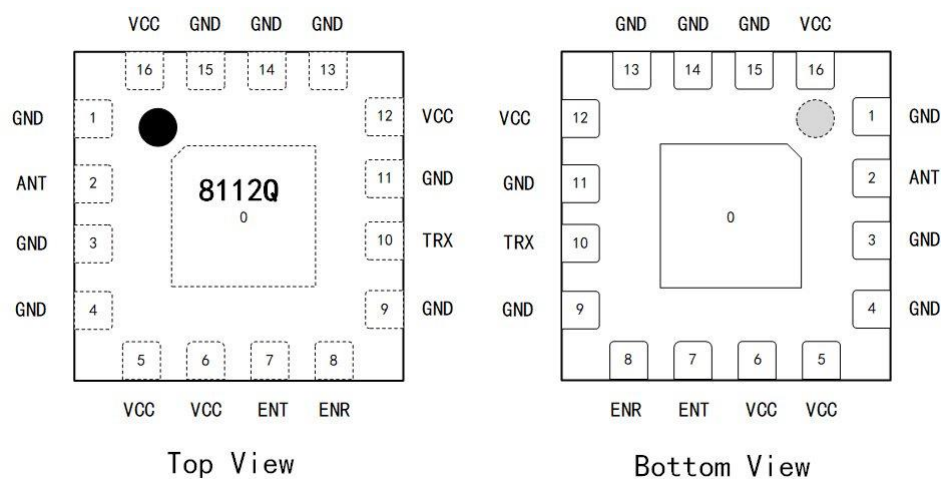


Figure 1. SW8112Q Pin Configuration and Marking

PIN NAMES AND SIGNAL DESCRIPTIONS

Table1. PIN name and description

Pin Num.	Name	Description
0	GND	GND pad
1	GND	GND pad
2	ANT	ANT pad
3	GND	GND pad
4	GND	GND pad
5	VCC	3.3V supply
6	VCC	3.3V supply
7	ENT	Enable of TX
8	ENR	Enable of RX
9	GND	Ground
10	TRX	TRX pad

11	GND	Ground
12	VCC	3.3V supply
13	GND	Ground
14	GND	Ground
15	GND	Ground
16	VCC	3.3V supply

ORDER INFORMATION

Table 2. Order Information

Part Number	Temperature	Package	RoHS	Mark	SPQ
SW8112Q	-40℃ ~ 85℃	3.0mm*3.0mmQFN-16L	Yes	8112Q	Tape and Reel 3000 pcs/Reel

TYPICAL APPLICATION

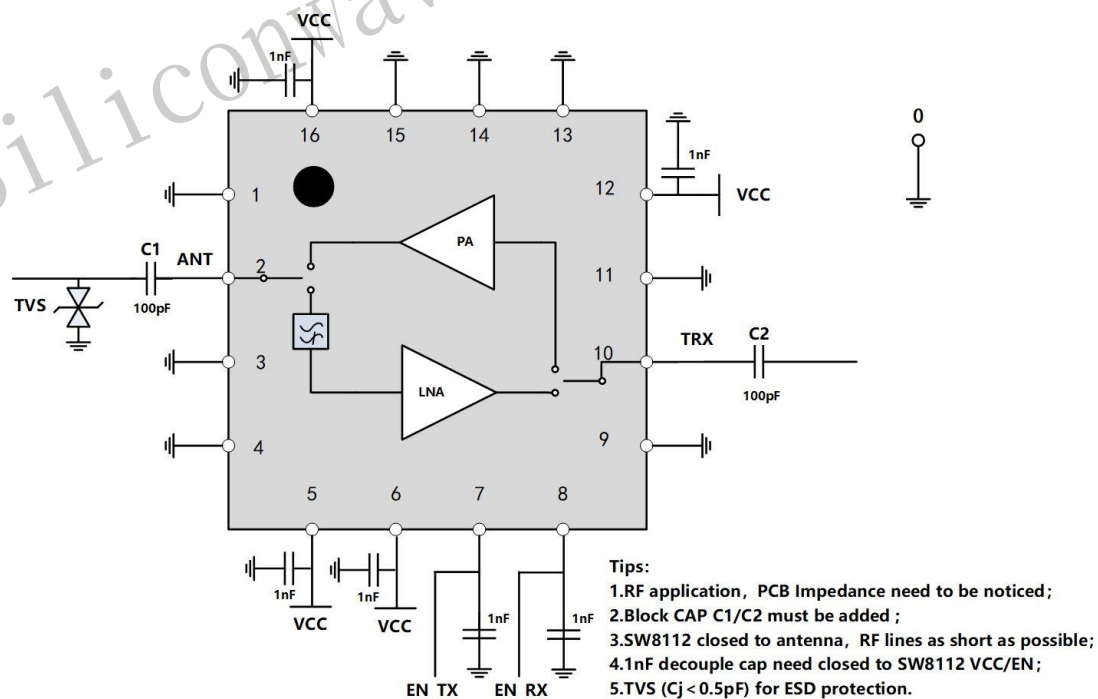


Figure 2. Typical Application Circuit

ABSOLUTE MAXIMUM RATINGS

Table 3. Limiting Values

Parameter	Symbol	Values			Unit
		Min.	Typ.	Max.	
Supply Voltage at Pin VCC	V _{CC}	-0.3	-	15.0	V
Voltage at Pin EN ²⁾	V _{EN}	-0.3	-	15.0	V
Current into Pin VCC	I _{CC}	-	-	300	mA
RF input Power ³⁾	P _{IN}	-	-	20	dBm
Package Thermal Resistance	θ _{JA}	-	TBD		°C/W
Junction Temperature	T _J	-	-	150	°C
Storage Temperature Range	T _{STG}	-65	-	150	°C
Ambient Temperature Range	T _{amb}	-40	-	85	°C
Solder Temperature(10s)		-	260	-	°C

ELECTRICAL CHARACTERISTICS

(SW8112Q EVB¹⁾; V_{CC}=2.8 to 3.6V, T_A=-40~+85°C, f=6000MHz to 9000MHz; Typical values are at V_{CC}=3.3V and T_{amb}=+25°C, f=7000MHz, unless otherwise noted.)

Table 4 .Electrical Characteristics

Parameter		Conditions	Min.	Typ.	Max.	Units
DC ELECTRICAL CHARACTERISTICS						
V _{CC}	Supply Voltage		2.8	3.3	3.6	V
V _{EN}	Digital logic High		2.8	3.3		V
V _{EN}	Digital logic LOW				0.45	V
I _{CC}	TX Supply Current	ENT=High; ENR=Low		40.0		mA
I _{CC}	RX Supply Current	ENT=Low; ENR=High		30.0		mA
I _{SD}	Shut Down Current	ENT=Low; ENR=Low		3.0		uA
I _{EN}	EN Supply Current			1.0		mA
TRANSMIT MODE AC ELECTRICAL CHARACTERISTICS						

Fre	Frequency		6.0	7.0	8.5	GHz
Gp _{TX}	TX Power Gain	RF=-35dBm	20.0	25.0	18.0	dB
P _{1dB, TX}	TX Output 1dB Compression Point	f=7000MHz; TX mode		+15.5		dBm
P _{sat, TX}	TX Output saturation point	f=7000MHz; TX mode		+18.0		dBm
I _{cc-Psat}	TX current @P _{out} =+18dBm	f=7000MHz; TX mode		90		mA
HARM	Harmonics	2f ₀ , 3f ₀			-40	dBm
ISO	Forward Isolation	ENR=Low; ENT=High		30.0		dB
RECEIVE MODE AC ELECTRICAL CHARACTERISTICS						
Fre	Frequency		6.0	7.0	8.5	GHz
NF _{RX}	RX Noise Figure			2.0	2.2	dB
RL _{in}	Input Return Loss			12.0		dB
PG	Power Gain		19.0	17.0	15.0	dB
RL _{out}	Output Return Loss			5.0		dB
P _{1dB, RX}	RX Output 1dB Compression Point			+10		dBm
REJ	Out-Of-Band Rejection	DC-2.4GHz		36		dBc
RECEIVE MODE AC ELECTRICAL CHARACTERISTICS						
T _{power on}	Power-on Time	Supply from low to high		5		μs
t _{on}	Turn-on Time	TX to RX		0.5		μs
t _{off}	Turn-off Time	RX to TX		0.5		μs

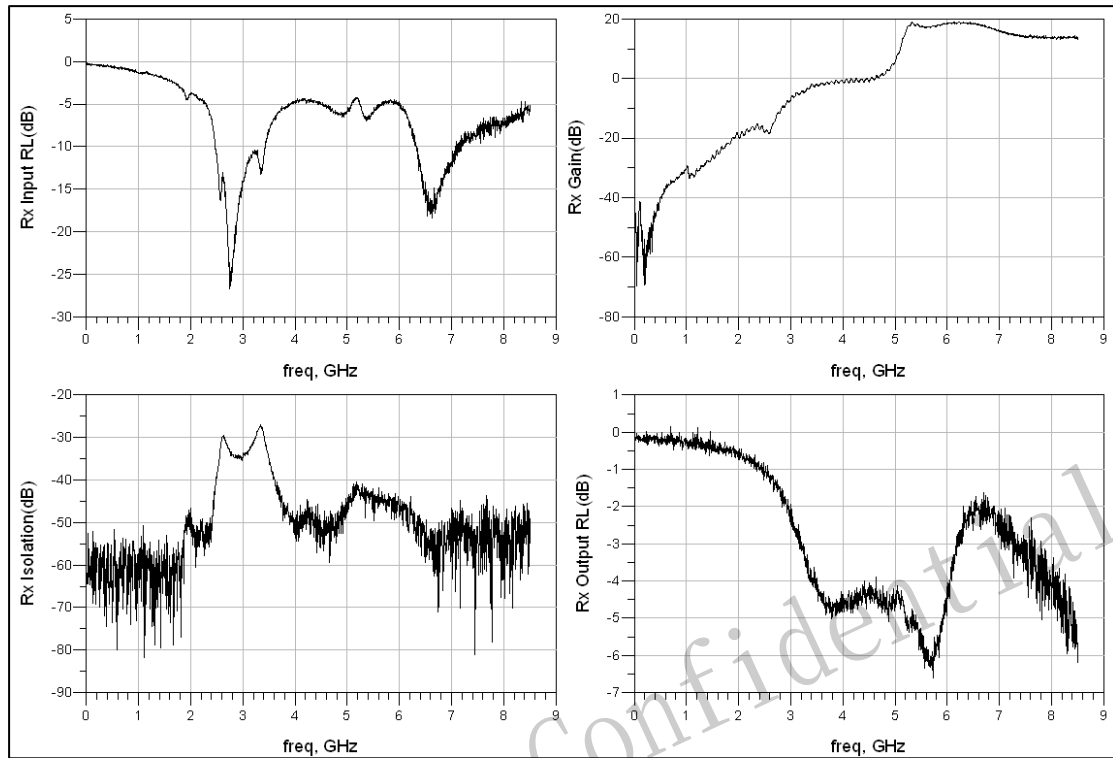
ENABLE CONTROL TABLE

Table 5. EN control table

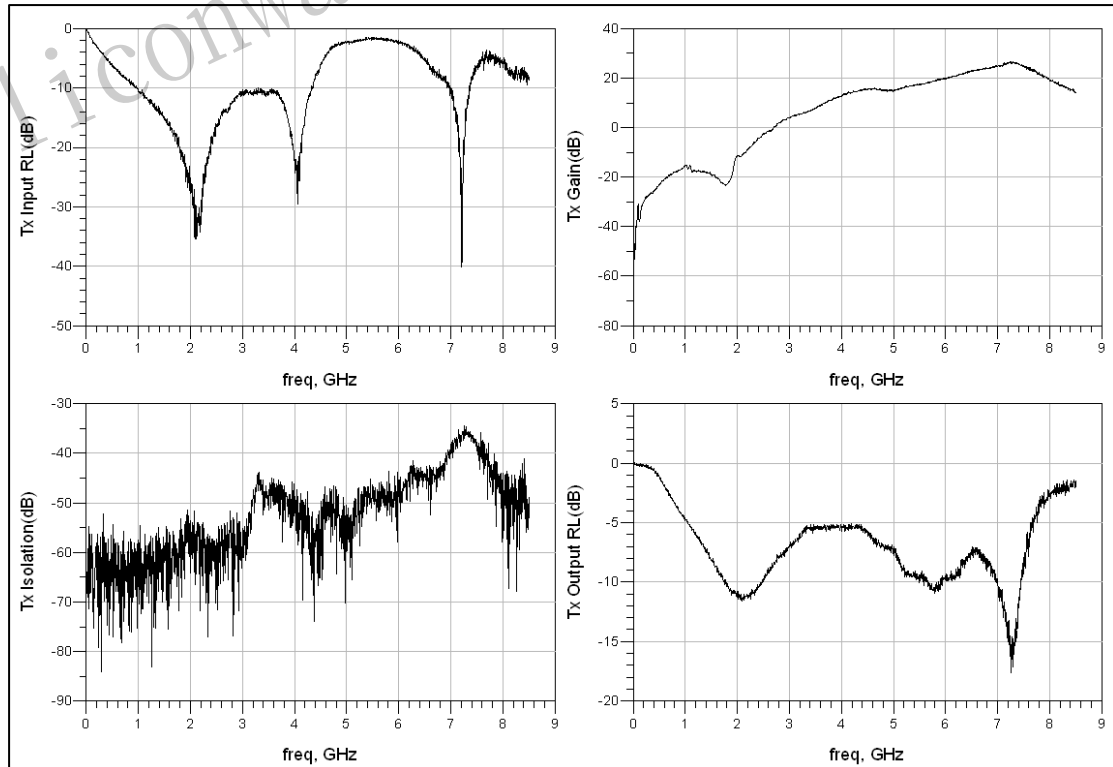
ENT	ENR	STATUS
HIGH	LOW	TX
LOW	HIGH	RX
LOW	LOW	STANDBY
HIGH	HIGH	FORBIDDEN

TYPICAL OPERATING GRAPHICS

RX S-PARA VCC=3.3V EN=3.3V SWEEP POWER=-30DBM



TX S-PARA VCC=3.3V EN=3.3V SWEEP POWER=-30DBM



1. The SW8112Q requires no external capacitor/inductor for input matching. Optional supply decoupling capacitor is 1nF.
2. The SW8112Q should be placed close to the antenna. Use 50-ohm micro-strip lines to connect RF INPUT and RF OUTPUT. Bypass capacitor should be located close to the device. For long supply lines, it may be necessary to add more decoupling capacitors. Proper grounding of the GND pins is very important.

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Figure 3. Package Outline

RECOMMENDED SOLDER TEMPERATURE

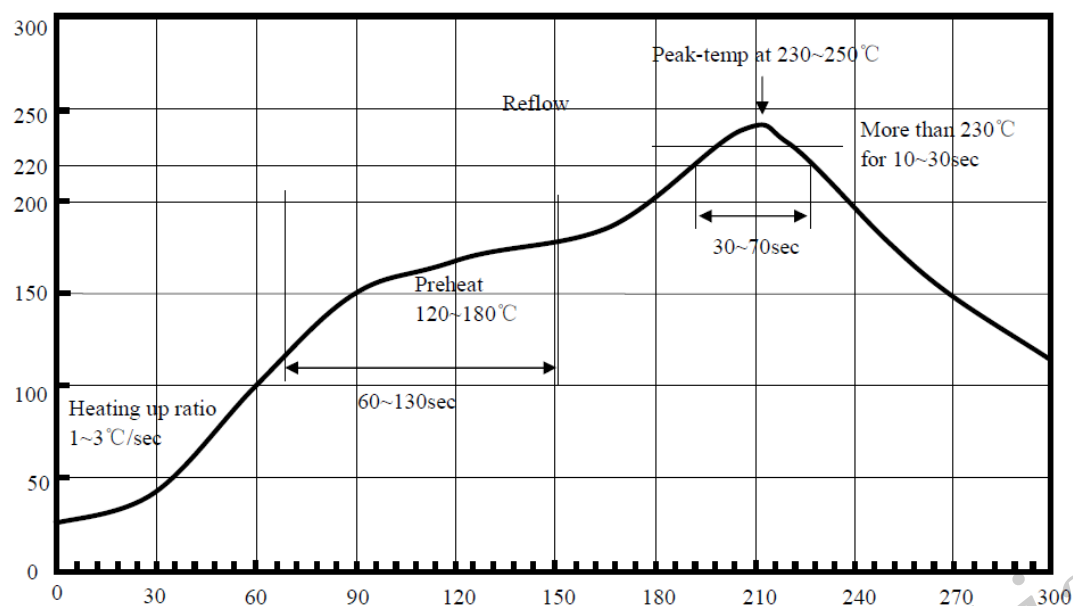


Figure 4. Recommended Solder Temperature

ROHS COMPLIANT

The product does not contain lead, mercury, cadmium, hexavalent chromium, polybrominated biphenyls(PBB) or polybrominated diphenyl ethers(PBDE), and are therefore considered RoHS compliant.

REVISION HISTORY

Document ID	Release Date	Change Record
V0.95	2020.07	Beta
V1.0	2020.08	Officially released